MILITARY SPECIFICATION

MOUNTING KIT: M1 TANK,

CLEARED LANE MARKING SYSTEM (CLAMS)

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of a mounting kit used in attaching the cleared lane marking system (CLAMPS) to the M1 series tank.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490	- Cleaning Method and Pretreatment of Ferrous Surfaces for Organic Coatings
PPP-B-601	- Boxes, Wood, Cleated-Plywood.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2590

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MIL-B-121	- Barrier Material, Grease Proofed, Waterproofed, Flexible.
MIL-B-5087	- Bonding, Electrical, and Lighting Protection, for Aerospace Systems.
MIL-P-14105	- Paint, Heat-Resisting For (Steel Surfaces).
MIL-C-16173	- Corrosion Preventative Compound, Solvent Cutback, Cold Application.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Tables and Procedures for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-280	- Definitions of Item Levels, Item Exchangeability, Models, and Related Terms.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-471	- Maintainability Demonstration.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-882	- System Safety Program Requirements.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing: with Appropriate Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-45662	- Calibration System Requirements.
MS 25083	- Jumper Assembly, Electric, Bonding.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified: the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ME

TA13228E2100	- Mounting Kit, CLAMS to M1 Series Tank.
13228E2030	- Converter Box Assembly.

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13228E2040	- Cover
13228E2045	- Panel Assembly, R.H.
13228E2048	- Panel Assembly, L.H.
13228E2052	- Panel Assembly, Back.
13228E2066	- Adapter Assembly.
12344343	- Interior CARC Paint Specification.
12344344	- Exterior CARC Paint Specification.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IV, Welding and Brazing Qualifications

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code - Steel

(Application for copies should be addressed to the American Welding Society, 550 N.W. Lejune Rd., P.O. Box 351040, Miami, FL 33135.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 General description. The M1 CLAMS mounting kit provides the interface between the CLAMS and the M1 series tank. The mounting kit shall consist of the following items: a structural adapter for mounting the CLAMS dispenser box at the rear of the vehicle, metallic heat insulation panels that will encase the CLAMS dispenser box and protect its contents from high temperature engine exhaust, a converter box to provide electrical interface between the vehicle and CLAMS control box which controls marker dispensation, and interconnecting electrical harnesses.

3.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances are permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the kit. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the procuring activity or his designated representative.

3.3 First article (first produced kit). Unless otherwise specified (see 6.2), one or more samples shall be subjected to first article inspection (see 4.3 and 6.3) to prove that his production methods will produce kits that comply with the requirements of this specification. Any changes or deviations of mounting kits from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish mounting kits conforming to this specification.

3.4 Performance characteristics.

3.4.1 Converter box.

3.4.1.1 Input power. Unless otherwise specified, the requirements specified herein are applicable when an input voltage within the range of 18 to 30 volts direct current (Vdc) is applied to terminals J1-K (positive) and J1-C (return) of the converter box.

3.4.1.2 Signal processing. For an input sine wave signal of 350 (+35, -0.0) millivolts peak-to-peak (mVp-p) to 4.0 ±1.0 volts peak-to-peak (Vp-p) between terminals J1-E and J1-F at frequencies of 50 ±5.0 to 1000 ±100 Hertz (Hz) the output signal between terminals J1-H (positive) and J1-C (return) shall be a pulse of 17.0 to 30.0 volts maximum amplitude and 0.3 ±0.2 volts minimum amplitude with a pulse duration of 2.0 ±1.0 milliseconds and a signal period of 13.06 (+1.45, -1.19) to 0.653 (+0.073, -0.059) (the converter shall divide the input signal frequency by 653.0).

3.4.1.2.1 Noise level. There shall be 0.3 ±0.2 Vdc present between the output terminals J1-H (positive) and J1-C (return) for an input signal between terminals Jl-E and Jl-F of 25 mVp-p or less for frequencies of 5 Hz to 10 KHz (10000 Hz).

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3.4.2 Structural adapter. The structural adapter shall be assembled in accordance with Drawing 13228E2066. The structural adapter when supported by the telescoping arms, and pintle mounting bar, shall support the mounting of the CLAMS dispenser box and heat insulation panels. The dispenser box shall pivot within the mounting clevis of the structural adapter with all heat insulation panels installed.

3.5 Design and construction. The mounting kit shall conform to the parts inventory list of 13228K2100.

3.5.1 Dimensions.

3.5.1.1 Structural adapter and heat insulation panels. The CLAMS structural adapter shall be fabricated in accordance with Drawing 13228E2066. The insulation panels shall be fabricated in accordance with Drawings 13228E2040, 13228E2045, 13228E2048, 13228E2052.

3.5.1.2 Converter box housing. The converter box housing shall conform to the requirements of Drawing 13228E2030.

3.5.2 Interchangeability. Interchangeability of mounting kit components shall be in accordance with MIL-STD-280.

3.5.3 Soldering. Unless otherwise specified, soldering of electrical connections shall be in accordance with MIL-STD-454.

3.5.4 Hardware. Mounting hardware shall be selected from AN, NAS, MS, or commercial standards.

3.5.5 Electromagnetic interference/electromagnetic compatibility (EMI/EMC).

3.5.5.1 Electromagnetic compatibility. The mounting kit, when installed in the vehicle in conjunction with the CLAMS and operated in all its modes of operation, shall not cause adverse effects to the vehicle or any other subsystems installed therein.

3.5.6 Bonds and grounds. Electrical bonding for equipment shall be designed in accordance with MIL-B-5087. Equipment mounted electrical connector to mounting surface shall not exceed 50 milliohms. Provisions shall be made for grounding equipment to the vehicle mounting surface. When grounding jumpers are employed, metal bonding strips having a maximum length-to-width ratio of 5 to 1 or braided wire jumpers equivalent to MS25083 shall be used. The dc bonding resistance across the enclosure to jumper junction shall not exceed 2.5 milliohms.

3.5.7 Reliability. The mounting kit shall have a mission reliability of 0.90.

3.5.8 Maintainability. The mounting kit shall not require preventive maintenance. The mean-time-to-repair (MTTR) at the organizational level of maintenance shall not exceed 1.0 clock hours.

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3.5.9 Weight. The weight of the mounting kit shall not exceed 413 pounds, including all mounting hardware and kit detail components.

3.5.10 Human factors engineering. The mounting kit shall conform to human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, paragraphs 4 (General Requirements), 5.5 (Labeling), 5.6 (Anthropometry), 5.9 (Design for Maintainability) and 5.13 (Hazards and Safety) of MIL-STD-1472, as applicable.

3.6 Environmental requirements.

3.6.1 Temperature.

3.6.1.1 Operating temperature. The mounting kit shall meet the requirements of 3.4 during and after exposure to temperatures in the range of -25 to +140 ºF.

3.6.1.2 Storage temperature. The mounting kit shall meet the requirements of 3.4 after exposure to temperatures in the range of -60 to +160 °F.

3.6.2 Vibration. The mounting kit shall meet the requirements of 3.4 during and after exposure to sinusoidal vibration in accordance with table I, in each of the three mutually perpendicular axes. The vibration shall consist of a logarithmic sweep rate of 15 minutes per sweep cycle form 5 to 500 to 5 hertz (Hz).

TABLE I. Vibration levels.

Axis	Frequency (Hz)	Amplitude
Vertical	5 to 25 25 to 57 57 to 500	±1.0 g 0.03 inch D.A. ±5.0 g
Latitudinal and Longitudinal	5 to 25 25 to 44 44 to 500	±1.0 g 0.03 inch D.A. ±3.0 g

3.6.3 Basic shock. The CLAMS mounting kit shall meet the requirements of 3.4 during and after exposure to shock impulses of 30 ±3 g, 11 ±1.1 milliseconds (ms) half sine wave applied in each direction of the three mutually perpendicular axes.

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3.6.4 Ballistic shock. The mounting kit shall meet the requirements of 3.4 during and after exposure to shock impulses of 200 ±20 g, 0.5 ±0.1 ms half sine wave applied in each direction of the three mutually perpendicular axes.

3.6.5 High intensity shock. The converter box and parts thereof, shall remain intact and shall not become secondary projectiles when subjected to high intensity shock levels of 980 ±98 g, 0.5 ±0.1 ms at the assembly's mounting interface. This shall include the impulses in each direction of the three mutually perpendicular axes.

3.6.6 Submergence. There shall be no evidence of air leakage when the converter box is pressurized with air to 6.0 ±0.5 psig and completely submerged in water to a depth of 6 inches minimum measured from top surface. Electrical connectors shall be capped during submergence. The CLAMS converter box shall meet the requirements of 3.4 after submergence.

3.6.7 Cleaning spray. The converter box shall not be damaged by and shall meet the performance requirements of 3.4 after exposure to a jet spray of tap water applied perpendicular to and at a distance of one (+1.0, -0.0) foot from the component surface at a cleaning rate of 1.0 square foot per minute for a period of 10 minutes minimum. The water jet shall be derived from nozzle having a orifice diameter of 0.25 inch and a nozzle pressure of 25 psig.

3.7 Treatment and painting.

3.7.1 External. The portions of the mounting kit, not including the deflector, normally painted shall be cleaned, treated, and painted in accordance with Drawing 12344344 color 383 green. The deflector shall be cleaned per applicable method of specification TT-C-490 and painted with MIL-P-14105 color black.

3.7.2 Internal. The portions of the mounting kit, internally located on the tank, normally painted, shall be cleaned, treated, and painted in accordance with Drawing 12344343 color 17925 untinted white.

3.8 Identification and markings. The mounting kit shall be identified and marked in accordance with MIL-STD-130.

3.9 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9. Workmanship shall be such that the assembly shall be free of burrs, sharp edges, cracks, corrosion, surfaces out of alignment, or out of contour (visually apparent). There shall be no evidence of conditions that could present a safety hazard to operating or maintenance personnel. The assembly shall contain no loose or foreign material.

3.9.1 Welding certification.

3.9.1.1 Welding. The surface of parts to be welded shall be free from rust, corrosion, scale, paint, grease, and mill scale that can be removed by chipping and wire brushing, as well as from other foreign matter. Welds shall withstand stress without permanent deformation, or failure, when the parts connected by the by the welds are subjected to proof and servicing loading. Parts to be joined by fillet welds shall be

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bought into close contact as possible, and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. The welding process used in fabrication of the mounting kit shall be as specified in applicable drawings. No deviations from these requirements are permissible without prior approval from the procuring activity.

3.9.1.2 Welders and welding operators. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall make available to the contracting officer certification that the welder or welding operator has passed qualification tests as prescribed, by either of the following listed codes, for the type of welding operations to be performed, and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code - Steel
ASME Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications

Contractors who make only horizontal welds need not qualify welders or welding operators for "all position welding." The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes. The Government reserves the right to require recertification of any welder or welding operator.

3.9.1.3 Inspection. Inspect all welds as per drawings. No cracks are permissible. Workmanship specimens that are required by the applicable specifications shall be approved by quality assurance representatives of the procuring activity.

3.10 Safety. The mounting kit shall be designed in accordance with the design guidelines of MIL-STD-882. Electrical components shall meet requirement 1 of MIL-STD-454.

3.11 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.11.1 Material deterioration prevention and control. The mounting kit shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the item may be exposed.

3.11.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.11.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with each component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

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3.11.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the mounting kit may be newly fabricated from recovered materials to the maximum extent practicable, provided the mounting kit produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the mounting kit.

3.12 Qualification. Qualification is required for new vendors and shall be performed in accordance with the first article inspection of 3.3 unless otherwise specified by the procuring activity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection of equipment. Unless otherwise specified in the contract, the supplier is responsible for the provision and maintenance of all inspection and test equipment necessary to assure that supplies and services conform to contract requirements. Commercial, modified commercial, or contractor designed inspection equipment or measuring set-ups must be capable of repetitive measurements to an accuracy of 10 percent of component tolerance. Calibration of inspection and test equipment shall be in accordance with MIL-STD-45662, notice 3.

4.2 Classification of inspection. The inspection specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

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4.3 First article inspection.

4.3.1 Examination. The mounting kit shall be examined as specified in 4.5.1 in the order shown. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The mounting kit shall be subjected to tests marked "X" in column 2 of the table in 4.5.2, in the order shown. Failure of any test shall be cause of rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5. AQL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.0 percent defective.

4.5 Examination procedure.

4.5.1 Examination. The mounting kit shall be examined as specified herein for the following defects:

MAJOR

101. Material not as specified.
102. Components missing, not as specified, or damaged.
103. Dimensions not as specified.
104. Weight not as specified.
105. Interchangeability not as specified.
106. Soldering not as specified.
107. Mounting hardware not as specified.
108. Human performance/human engineering not as specified.
109. Welding not as specified.
110. Certification of welders not as specified.
111. Safety not as specified.
112. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments as specified.
113. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other as specified.
114. Contractor does not have documentation available for identification of material, material finishes, or treatment.
115. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the mounting kit.

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201. Treatment and painting not as specified.
202. Identification and marking not as specified.
203. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test schedule.

TABLE II. Test schedules.

Test Order	First Article	Quality Conform	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
1	X	X	Input power	4.5.3.1	3.4.1.1
2	X	X	Signal processing	4.5.3.2	3.4.1.2
3	X	X	Noise level	4.5.3.3	3.4.1.2.1
4	X	X	Structural adapter	4.5.3.4	3.4.2
5	X	X	Electromagnetic compatibility	4.5.4	3.5.5.1
6	X	X	Bonds and grounds	4.5.5	3.5.6
7	X		Reliability	4.5.6	3.5.7
8	X		Maintainability	4.5.7	3.5.8
9	X		Low temperature	4.5.8.1	3.6.1.1
10	X	X	High temperature	4.5.8.2	3.6.1.1
11	X		Storage temperature	4.5.8.3	3.6.1.2
12	X		Vibration	4.5.9	3.6.2
13	X		Basic shock	4.5.10	3.6.3
14	X		Ballistic shock	4.5.11	3.6.4
15	X		High intensity shock	4.5.12	3.6.5
16	X	X	Submergence	4.5.13	3.6.6
17	X		Cleaning spray	4.5.14	3.6.7

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4.5.3 Performance characteristics.

4.5.3.1 Input power. Unless otherwise specified, perform the tests specified herein as 18.0 (+0.5, -0.0) Vdc and again at 30.0 (+0.0, -0.5) Vdc. Apply input power to the terminals specified in 3.4.1.1 and verify assembly performance.

4.5.3.2 Signal processing. As specified in 3.4.1.2 apply an input signal of 350 (+35, -0.0) mVp-p at a frequency of 50.0 (±5.0) Hz between J1-E and J1-F. Measure and verify the output at terminals J1-H (positive) and J1-C (return) has the following characteristics:

1. Maximum pulse amplitude of 17.0 to 30.0 Vdc and minimum pulse amplitude of 0.3 ±0.2 Vdc.
2. The duration of the maximum amplitude portion of output signal shall be ±1.0 milliseconds.
3. The period of the output signal shall be 13.06 (+1.45, -1.19) seconds.

Repeat the above test with an input frequency of 1000 ±100 Hz in place of the 50.0 Hz.

NOTE: for an input frequency of 1000 ±100 Hz the period (Item 3) shall be 0.653 (+0.726, -0.594) seconds.

Repeat the complete sequence of tests with an input signal of 5.0 (+0.0, -0.5) Vp-p in place of the 350.0 mVp-p signal.

4.5.3.3 Noise level. Apply an input signal of 25.0 ±2.5 mVp-p with frequency of 3000 ±300 Hz between J1-E and J1-F. Measure and verify the output voltage at J1-H (positive) and J1-C (return) is in conformance with 3.4.1.2.1.

NOTE: This test is to be performed on the 100 percent acceptance tests. The control test for this section shall address the frequencies of 3.4.1.2.1.

4.5.3.4 Structural adapter. Support the structural adapter by the telescoping hull lifting eye mounting tubes and pintle mounting bar. Verify that the dispenser box pivots within the mounting clevis of the structural adapter with heat insulation panels installed, as specified in 3.4.2.

3.5.4 Electromagetic compatibility. Compliance to 3.5.5.1 shall be determined by conducting an intra system EMG test on an M1 tank with CLAMS and CLAMS mounting kit installed.

4.5.5 Bonds and grounds. Measure the dc resistance, in accordance with MIL-STD-202, method 303, across each bond ground junction with interfacing harnesses disconnected to verify that the dc resistance is as specified in 3.5.6. Visually verify that the bondings are installed in accordance with 3.5.6. Verify the jumpers are as specified in 3.5.6.

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4.5.6 Reliability. The mounting kit shall be tested as follows:

4.5.6.1 Test conditions. Unless otherwise specified, the reliability shall be conducted under the following conditions:

Air temperature	73 ±18 °F
Barometric pressure	28.5 (+2.0, -3.0) inches of mercury
Relative humidity	50 ±30 percent

4.5.6.2 Accept and reject criteria. The mounting kit shall complete twenty two (22) missions with zero (0) relevant failures. Failures shall be determined by the Government (see 4.5.6.3 for failure definition and 4.5.6.4 for test procedure).

4.5.6.3 Failure definition and scoring criteria. A failure is defined as any malfunction which the operator cannot remedy in 2 minutes or less and which causes any or all of the following: The CLAMS dispenser not to be held in a position for proper operation; the CLAMS not to be properly operated by the crew inside the tank; the CLAMS dispenser not to be protected from the heat of exhaust; the mounting kit and the CLAMS to be mounted in more than 30 minutes. Any failures attributed to the CLAMS or the M1 tank will not be chargeable to the mounting kit. Pattern failures are considered one failure. Failures will be determined by the Government.

4.5.6.4 Test procedure. A mission for the mounting kit will be as follows:

a. The M1 tank shall rendezvous with the support vehicles at the beginning of the mission.
b. The mounting kit and the CLAMS shall be installed on the M1 tank within 30 minutes.
c. The M1 tank shall then conduct a mine breach of 100 meters over varying terrains. The CLAMS shall be operated during each mission. The automatic dropping of markers shall be different for each mission so that all possible distances are tested at least twice.
d. After the mine field breach is completed, the CLAMS dispenser shall be shut off, and for the duration the 45 minutes operational mode, which started at the beginning of the mine field breach, the M1 tank shall traverse secondary roads at varying speeds up to 10 mph.
e. The position and orientation at the markers dispensed shall be recorded, the distance of each marker from the centerline of the breaching tank and whether the marker is in an upright position. The distance between each marker shall be recorded as well as the setting of the automatic dispensing. All maintenance actions, and times shall be recorded.

4.5.7 Maintainability. The mounting kit shall be tested in accordance with MIL-STD-471 and as specified herein, utilizing test method 1 for mean-time-for repair requirement.

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4.5.8 Operating temperature.

4.5.8.1 Low temperature. Subject the mounting kit to the low temperature test specified in MIL-STD-810, method 502.1, procedure 1 to verify conformance to 3.6.1.1. Maintain the temperature to -6 °F for a period of 24 hours. At the conclusion of this time, stabilize the assembly at the low extreme temperature specified in 3.6.1.1 and perform the test of 4.5.3. Return the assembly to 73 ±18 °F and perform the tests of 4.5.3.

4.5.8.2 High temperature. Subject the mounting kit to the high temperature test specified in MIL-STD-810, method 501.1, procedure 1 to verify conformance to 3.6.1.1. Maintain the temperature at 160 °F for a period of 48 hours. At the conclusion of this time, stabilize the assembly at the high extreme temperature specified in 3.6.1.1 and perform the tests of 4.5.3. Return the assembly to 73 ±18 °F and perform the tests of 4.5.3.

4.5.8.3 Storage temperature. The mounting kit is to be tested in accordance with the tests of 4.5.8 at the temperature extremes specified in 3.6.1.2 to verify conformance to 3.6.1.2.

4.5.9 Vibration. Mount the mounting kit, using the installed interface configuration for its intended application, or equivalent, and subject it to the test of MIL-STD-810, method 514.2, procedure V111 to verify conformance to 3.6.2. Subject the assembly to a sinusoidal vibration test in accordance with the test level requirements of table I, for a total 180 minutes, in each of three mutually perpendicular axes. The vibration is to be imposed at a logarithmic sweep rate of 15 minutes per sweep which cycling between 5 to 500 to 5 Hz (during exposure of converter box, monitor output signal to verify that intermittent operation does not occur). At the conclusion of this test, examine the assembly for damage and subject it to the performance test of 4.5.3.

4.5.10 Basic shock. Verify conformance to 3.6.3 with the mounting kit mounted, using the installed interface configuration for its intended application, or equivalent, and subject it to the test of MIL-STD-810, method 516.2, procedure I. Apply a total of three half sine wave shock pulses in both directions along each of three mutually perpendicular axes for a total of 18 shocks. With peak amplitude in accordance with 3.6.3 (during exposure of the converter box), monitor the output signal of the converter box to verify that intermittent operations does not occur. At the conclusion of this test, examine the assembly for damage and subject it to the performance test of 4.5.3.

4.5.11 Ballistic shock. Mount the mounting kit, using the installed interface configuration for its intended application, or equivalent, and subject it to the test of MIL-STD-810, method 516.2, procedure VI To verify conformance with 3.6.4. Apply to total of three half sine wave shock pulses in both directions along each of three mutually perpendicular axes for a total of 18 shocks. Peak amplitude in accordance with 3.6.4 (during exposure of the converter box), monitor converter box output signal to verify that intermittent operation does not occur). At the conclusion of this test, examine the assembly for damage and subject it to the performance tests of 4.5.3.

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4.5.12 High intensity shock. Mount the mounting kit, using the installed interface configuration for its intended application, or equivalent, and subject it to the test of MIL-STD-810, method 516.2, procedure VI to verify conformance with 3.6.5. A total of three half sine wave shock pulses is to be applied in both directions along each of three mutually prependicular axes for a total of 18 shocks. Peak amplitude is to be in accordance with 3.6.5. During exposure the assembly is to be monitored visually to verify it remains intact and that no parts of the assembly become secondary projectiles.

4.5.13 Submergence. To verify conformance of 3.6.6 perform the submergence test as follows:

Install seal caps to all electrical connectors and pressurize the assembly with air through a pressure adapter to 6.0 ±0.5 psig. Submerge the assembly in a suitably-sized container of clear water to a depth of 6.0 (±1.0, -0.0) inch (measure to verify submergence depth between the water surface and the top surface of the assembly) for a minimum period of 5 minutes. Observe the assembly for leakage as evidenced by air bubbles escaping from the interior of the assembly. Any indication of leakage is cause for rejection. Remove the assembly from the water prior to venting the assembly pressure to zero psig. After exposure to the submergence test, subject the assembly to the performance tests of 4.5.3.

4.5.14 Cleaning spray. Subject the mounting kit to a cleaning spray test as specified in 3.6.7. After exposure to the cleaning spray, the assembly is to be subjected to the performance tests of 4.5.3.

4.5.15 Safety. The mounting kit shall be evaluated for safety requirements throughout testing as specified herein. Nonconformance to 3.10 shall constitute failure of this test.

4.5.16 Human factors engineering. The mounting kit shall be evaluated by a qualified human factors engineer. Nonconformance to 3.5.10 shall constitute failure of this inspection.

4.6 Inspection of packaging.

4.6.1 First article packaging inspection. The first article pack shall be examined for the defects listed in 4.6.2.3. The presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of packaging.

4.6.2.1 Unit of product. For the purpose of inspection only, a unit of product shall be a complete mounting kit that has been prepared for shipment.

4.6.2.2 Sampling. Sampling for inspection shall be in accordance with MIL-STD-105.

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4.6.2.3 Examination. Samples selected for examination in accordance with 4.6.2.2 shall be examined for the following defects. The AQL shall be 1.0 percent defective.

No.	Defect	Level A	Level C
116.	Unpainted or machined ferrous metal surfaces not preserved as specified.	5.2.1	
117.	Items in carton not wrapped as specified.	5.2.1	
118.	Exposed ends of hydraulic lines not protected as specified.	5.2.2	
119.	Technical publications not sealed in polyethylene bag as specified.	5.2.3	
120.	Shipping container not as specified.	5.3.1	5.3.2
121.	Strapping not as specified.	5.3.1	5.3.2
122.	Contents not cushioned, blocked, braced and anchored within shipping container as specified.	5.3.1	5.3.2
123.	Marking not as specified.	5.4	5.4

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the frame specified (see 6.2), to prove, prior to starting production packaging, that the applied preservation, packing and marking comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.6). The first article pack may be accomplished utilizing either the first article model or a production model. If the first article model is utilized, and the Government requests a comparison between the first article model and a production model, any preservation and packing shall be removed by the contractor at no expense to the Government

5.2 Preservation.

5.2.1 Ferrous metal surfaces. Any unpainted or machined ferrous metal surfaces on any of the mounting kit components shall be coated with preservative conforming to MIL-C-16173, grade 2, or equivalent. Those items placed in individual cartons shall be wrapped with barrier material conforming to MIL-B-121.

5.2.2 Hydraulic lines. Hydraulic lines, both rigid and flexible types, not connected to another component, shall have their ends sealed with plastic caps or plugs to prevent entrance of dirt or water.

5.2.3 Technical publications. Technical publications shall be placed together in a suitably sized polyethylene bag, not less than 4 mils, thick. The bag shall be closed by heat sealing.

5.3 Packing. Packing shall be level A or C as specified (see 6.2).

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5.3.1 Level A. Each complete mounting kit, preserved as specified in 5.2, shall be packed as a kit in a snug-fitting box conforming to PPP-B-601, overseas type, style A. Box closure and strapping shall be as specified therein except that the strapping shall be flat and the finish "B". The contents shall, as applicable to prevent damage to the contents or box, be cushioned, blocked, braced and anchored in accordance with the requirements of MIL-STD-1186.

5.3.2 Level C. Each complete mounting kit, preserved as specified in 5.2, shall be packed for level C in the same manner as specified for level A in 5.3.2 except that the box shall be domestic type and the strapping may be finish "A".

5.4 Marking. In addition to any special or identification markings required by the contract or purchase order, each container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The M1 CLAMS mounting kit permits the mounting of the CLAMS dispenser box and attachment of CLAMS control box and control cable on the Ml series tank.

6.2 Ordering data. The basic acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable to this contract and exceptions thereto.
c. Time frame required for submission of first article kit (see 3.3).
d. Data requirements.
e. Government and contractor inspection responsibilities (see 4.1).
f. Time frame required for submission of welding workmanship specimens (see 3.9.1.3).
g. Degree of packing required (see 5.3).
h. Number of units required for first article inspection.
i. Time frame for submission of first article pack (see 5.1).
j. Any special markings (see 5.4).

6.3 First article. When a first article inspection is required, the items should be a first produced model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Vehicle performance. Installation of the mounting kit and CLAMS on M1 series vehicles may degrade vehicle performance in the followings areas:

1. Electromagnetic interference (EMI).
2. Infrared signature.
3. Nuclear survivability.

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4. Mobility, i.e., 60 percent incline in reverse, vehicle power response and range, and vehicle towing capability.
5. Power pack removal.

6.5 Level of preservation. For the purpose of marking, the preservation specified in 5.2 shall be designated as "level A".

6.6 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack and mark the tanks in accordance with this specification.

6.7 Subject term (key word) listing.

Cleared lane marking system (CLAMS), M1
Mounting Kit, CLAMS
Mounting kit, M1

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
Army - AT

Project 2590-A008

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